

Problem 3.3

Assume that $A(t) = 100 + 5t$.

(a) Find i_5 .

(b) Find i_{10} .

Solution

(a) By definition,

$$i_5 = \frac{A(5) - A(4)}{A(4)}.$$

Since

$$A(5) = 100 + 5(5) = 125, \quad A(4) = 100 + 5(4) = 120,$$

we get

$$i_5 = \frac{125 - 120}{120} = \frac{5}{120} = 0.041\bar{6}.$$

Therefore,

$$i_5 \approx 4.16\%.$$

(b) Similarly,

$$i_{10} = \frac{A(10) - A(9)}{A(9)}.$$

Since

$$A(10) = 100 + 5(10) = 150, \quad A(9) = 100 + 5(9) = 145,$$

we obtain

$$i_{10} = \frac{150 - 145}{145} = \frac{5}{145} \approx 0.03448.$$

Therefore,

$$i_{10} \approx 3.448\%.$$